

Conceptual Soundness Validation Report

Model Name: SMI3.0 Mortgage Insurance Credit Risk Model

1. Model Purpose

Objective

To assess whether the model objective, target definition, and intended use are clearly defined and aligned with the business purpose.

Description

Technical Model Validation reviewed the model objective, intended use, target definition, and overall scope to determine whether the model is appropriate for mortgage insurance credit risk assessment and underwriting decisions. The assessment also considered whether the modelling framework is aligned with the intended business process and portfolio.

Validation Assessment

Review Item	Status
Model objective clearly defined	Yes
Model use case clearly identified	Yes
Target and decision point are consistent	Yes
Model scope appropriate for portfolio	Yes

Validator Comments

Technical Model Validation confirmed that the SMI3.0 model is designed to support mortgage insurance credit risk assessment and underwriting decisions.

The modelling framework combines borrower, policy, property, credit bureau, and economic information to estimate mortgage credit risk at the application level. The target definition and intended business use are consistent with the overall objective of supporting underwriting decisions.

Based on the documentation and implementation reviewed, the model objective is clearly defined and appropriate for the intended mortgage insurance portfolio.

2. Risk Theory and Business Logic

Objective

To assess whether the model structure and risk relationships are consistent with established mortgage credit risk principles.

Description

The validation reviewed whether the selected risk drivers, feature construction, and modelling approach appropriately represent borrower affordability, collateral quality, credit characteristics, and economic conditions relevant to mortgage insurance risk.

Validation Assessment

Review Item	Status
Model structure consistent with credit risk theory	Yes
Risk relationships economically intuitive	Yes
Mortgage insurance risk drivers appropriately reflected	Yes
Direction of key risk relationships sensible	Yes

Validator Comments

The model incorporates key mortgage credit risk drivers that are commonly used in mortgage underwriting, including borrower credit quality, affordability measures, collateral characteristics, and regional economic conditions.

The selected variables and feature construction are consistent with established mortgage credit risk principles. Higher borrower leverage, weaker credit quality, and less favourable economic conditions are expected to increase mortgage default risk, while stronger borrower and collateral characteristics are expected to reduce risk.

Overall, the conceptual relationship between model inputs and expected outcomes is considered reasonable and aligned with industry practice.

3. Model Assumptions

Objective

To assess whether the key modelling assumptions are reasonable, documented, and appropriate for the intended model use.

Description

Technical Model Validation reviewed assumptions related to borrower aggregation, missing-value treatment, feature construction, portfolio segmentation, and use of economic variables.

Validation Assessment

Review Item	Status
Key assumptions documented	Partial
Assumptions reasonable and defensible	Yes
Assumptions consistent with portfolio behaviour	Yes
Simplifying assumptions identified	Partial
Assumptions suitable under stressed conditions	Partial

Validator Comments

The review identified several operational assumptions supporting the modelling framework, including borrower aggregation, missing-value treatment, credit score fallback logic, and economic feature integration.

These assumptions are generally reasonable for mortgage insurance underwriting and are consistent with the intended business purpose.

However, additional documentation would improve transparency around selected thresholds, fallback logic, and the timing of economic variables. Stress-period assumptions should also be documented more clearly to support future model governance.

4. Inputs and Variables

Objective

To assess whether the selected input variables are relevant, consistently defined, and available at the intended decision point.

Description

Technical Model Validation reviewed whether the model inputs appropriately support mortgage insurance credit risk assessment without relying on future information or variables that may introduce unintended target leakage.

Validation Assessment

Review Item	Status
Input variables relevant to objective	Yes
Variables available at decision point	Partial
Variables consistently defined	Yes
No unintended proxy outcome variables identified	Partial
No unavailable future information used	Partial

Validator Comments

The model uses information relating to borrower credit quality, affordability, property characteristics, regional factors, and macroeconomic conditions. These variables are appropriate for mortgage insurance credit risk assessment and underwriting decisions.

Technical Model Validation did not identify any obvious conceptual issues with the selected input variables. However, forward-looking economic variables should continue to be reviewed to confirm that only information available at the underwriting date is used during model development and prediction.

Overall, the selected variables are considered appropriate for the intended modelling objective.

This completes **Sections 1–4** in the same concise style as the previous validation report. The next part will cover:

- **5. Model Methodology**
- **6. Feature Engineering and Transformations**
- **7. Model Limitations**
- **8. Sensitivity and Plausibility**

- 9. Expert Judgment
- 10. Overall Conclusion and Required Actions

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Great. Below is the second half of the **Conceptual Soundness Validation Report**, written in the same concise style as the previous SMI2.0 validation report and based on the implementation and supporting artifacts you provided.

5. Model Methodology

Objective

To assess whether the selected modelling methodology is appropriate for the intended business use and available data.

Description

Technical Model Validation reviewed the overall modelling approach, including preprocessing, borrower aggregation, feature engineering, model training, and evaluation, to determine whether the methodology is suitable for mortgage insurance credit risk assessment.

Validation Assessment

Review Item	Status
Methodology appropriate for business problem	Yes
Model form consistent with data structure	Yes
Model complexity justified	Yes
Segmentation approach logical	Yes
Methodology aligned with industry practice	Yes

Validator Comments

The modelling methodology is appropriate for the intended mortgage insurance credit risk application.

The framework applies structured preprocessing, borrower-level aggregation, feature engineering, model training, and evaluation using a controlled workflow. The segmentation approach is logical and supports application-level risk assessment.

The overall model complexity is appropriate for the available data and intended business use. No material conceptual concerns were identified with the selected modelling methodology.

6. Feature Engineering and Transformations

Objective

To assess whether engineered features preserve economic meaning and improve the model without introducing unnecessary complexity or data leakage.

Description

Technical Model Validation reviewed the engineered variables and transformations used during model development to determine whether they appropriately represent borrower and collateral risk characteristics while maintaining model interpretability.

Validation Assessment

Review Item	Status
Engineered features economically meaningful	Yes
Transformations preserve risk relationships	Yes
Binning and scaling justified	Yes
Feature construction avoids leakage	Partial
Feature engineering reproducible and transparent	Yes

Validator Comments

The feature engineering framework includes variables representing borrower affordability, property characteristics, credit quality, and regional economic conditions. These features are relevant to mortgage insurance credit risk assessment and support the intended underwriting objective.

The transformations reviewed generally preserve the economic interpretation of the original variables and improve consistency across the modelling dataset.

No material conceptual concerns were identified. However, additional validation of potential data leakage associated with time-dependent economic variables would strengthen the overall feature engineering framework.

7. Model Limitations

Objective

To assess whether known model limitations have been identified and appropriately documented.

Description

Technical Model Validation reviewed the modelling framework to identify operational, methodological, and data-related limitations that could affect model performance or interpretation.

Validation Assessment

Review Item	Status
Model limitations documented	Partial
Known weaknesses identified	Partial
Scope limitations understood	Yes
Model use restricted to intended purpose	Yes
Residual risk acknowledged	Partial

Validator Comments

The review identified several areas where documentation can be improved.

These include documentation of modelling assumptions, preprocessing thresholds, feature stability, stress-period representativeness, and formal leakage assessment.

These observations are considered governance improvements rather than conceptual weaknesses and do not materially affect the overall suitability of the modelling framework.

8. Sensitivity and Plausibility

Objective

To assess whether the model behaves reasonably when key risk factors change.

Description

Technical Model Validation reviewed whether the model responds to changes in borrower, property, and economic characteristics in a manner consistent with mortgage credit risk theory.

Validation Assessment

Review Item	Status
Model responds appropriately to input changes	Yes
Sensitivity to key drivers plausible	Yes
Stressed scenarios produce reasonable behaviour	Partial
No unstable or unreasonable outputs identified	Yes
Model behaves consistently across segments	Partial

Validator Comments

The overall model behaviour is consistent with expected mortgage credit risk relationships.

Higher borrower leverage, weaker credit quality, and adverse economic conditions are expected to increase model risk, while stronger borrower and collateral characteristics are expected to reduce risk.

Additional stress testing and sensitivity analysis across different portfolio segments would provide further evidence supporting model robustness.

9. Expert Judgment

Objective

To assess whether expert judgment incorporated into the modelling framework is appropriate, documented, and adequately governed.

Description

Technical Model Validation reviewed business rules, thresholds, fallback logic, and other expert judgement used during data preparation and feature engineering.

Validation Assessment

Review Item	Status
Expert judgment documented	Partial
Expert judgment supported by business rationale	Yes
Overrides and subjective choices controlled	Partial
Expert judgment does not dominate the model	Yes

Validator Comments

The framework includes business rules supporting borrower aggregation, credit-score fallback logic, property value treatment, and regional grouping.

These rules are appropriate for mortgage insurance underwriting and are applied consistently throughout the modelling process.

Additional documentation describing the rationale for selected thresholds and business-rule decisions would strengthen governance and improve auditability.

10. Overall Conclusion

Validation Assessment

Conclusion Item	Status
Model conceptually sound	Yes
Model conceptually sound subject to remediation	Yes
Model not conceptually sound	No

Conclusion

Technical Model Validation reviewed the conceptual design of the SMI3.0 Mortgage Insurance Credit Risk Model, including the model objective, methodology, assumptions, input variables, feature engineering, and underlying risk logic.

The review confirmed that the modelling framework is appropriate for its intended mortgage insurance credit risk use. The selected methodology, input variables, and engineered features are consistent with established mortgage credit risk principles and support application-level underwriting decisions.

No material conceptual weaknesses were identified during the review.

Several opportunities were identified to strengthen model governance, including additional documentation of assumptions, feature engineering decisions, business-rule thresholds, stress testing, and data leakage assessment. These recommendations are intended to improve transparency and governance and are not expected to materially affect the conceptual validity of the model.

Overall, Technical Model Validation concludes that the SMI3.0 Mortgage Insurance Credit Risk Model is conceptually sound for its intended use, subject to completion of the recommended documentation and governance improvements.

Key Findings

- The model objective and intended use are clearly defined.
- The modelling approach is consistent with established mortgage credit risk principles.
- The selected risk drivers appropriately represent borrower, property, and economic risk.
- The feature engineering approach is economically meaningful and supports the modelling objective.
- No material conceptual issues were identified during the review.
- Additional documentation of assumptions, business rules, feature stability, stress testing, and leakage assessment is recommended.

Required Actions

Action	Priority	Owner
Document key modelling assumptions	Moderate	Model Development

Action	Priority	Owner
Complete formal data leakage assessment	High	Validation Team
Expand stress testing and sensitivity analysis	Moderate Model Risk	
Improve documentation of business rules and thresholds	Moderate Model Development	
Document feature stability assessment	Moderate Model Development	